

# AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

**08** Live sessions**90** min Per session**04** weeks Course duration**Runs 100% on Google Colab**

Nothing to install, works even on a Chromebook

**Built for beginners**

No prior Python or machine learning required

**Live code-along on real data**

Concrete, geotechnical, water and transport datasets

**Notebooks, assignments & solutions**

Plus a certificate and session recordings

**INSTRUCTOR****Dr. Abdul Waris**

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D., IIT Hyderabad



Research Excellence Award 2026, IIT Hyderabad



Invited talks at Oregon State University (USA), Norway & Spain

## Course outline

Problem → Data → Model → Validate → Interpret → Decide

**01 Python for civil engineers**

AI/ML basics, Colab, NumPy & pandas on concrete data

**02 From raw data to ML-ready**

Explore, clean & split data; catch outliers and leakage

**03 Your first prediction model**

Linear regression, error metrics & cross-validation

**04 Trees & ensemble models**

Decision trees, Random Forest & boosting for nonlinear data

**05 Classification for decisions**

Logistic regression, confusion matrix, precision & recall

**06 Artificial neural networks**

Architecture, activations, loss, epochs & overfitting

**07 Explainable AI & engineering judgement**

Feature importance & SHAP on a geotechnical case study

**08 Uncertainty & reliability analysis**

Monte Carlo with ML surrogates to estimate failure probability

**WHO SHOULD JOIN**

B.Tech & M.Tech students / PhD scholars / Faculty  
Practising engineers / Complete beginners to AI & ML

STARTS · LIMITED SEATS



**Saturday,  
17 October 2026**

Scan to register · [qudelastudios.com](https://qudelastudios.com) →



# Most ML courses start with algorithms. This one starts with engineering problems.

Every session follows the workflow you would use on a real project, so every line of code has an engineering purpose.

|                                                                                     |                                                                                     |    |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----|
|   | <b>Define the problem</b><br>Which mix-design inputs drive concrete strength?       | 01 |
|  | <b>Understand the data</b><br>Explore, clean and check it before you trust it.      | 02 |
|  | <b>Build a model</b><br>Regression, trees, ensembles and neural networks.           | 03 |
|  | <b>Validate it</b><br>Test on data the model has never seen.                        | 04 |
|  | <b>Interpret it</b><br>Does the model respect engineering physics?                  | 05 |
|  | <b>Decide responsibly</b><br>Account for uncertainty before acting on a prediction. | 06 |

EXAMPLES FROM

-  Concrete materials
-  Geotechnical
-  Transportation
-  Water & environment
-  Buildings & energy

# Built for civil engineers at every stage.

Whether you are still studying or already on site, you will leave able to apply ML to your own engineering problems.



**UG students**  
B.Tech / B.E. in civil engineering



**PG students**  
M.Tech / M.E. in any civil specialisation



**PhD scholars**  
and early-career researchers



**Faculty members**  
bringing data-driven methods into teaching



**Practising engineers**  
structural, geotech, transport, water



**Complete beginners**  
never written a line of Python? Perfect.

 **What you need**

- ✓ Basic civil engineering concepts
- ✓ Comfort with everyday engineering calculations
- ✓ A Google account and a web browser

 **What you don't**

- ✗ ~~Prior Python or programming~~
- ✗ ~~Prior machine learning experience~~
- ✗ ~~Advanced mathematics~~
- ✗ ~~Anaconda, MATLAB or any installs~~

# By the end, you'll be able to...

|                                                                                     |                                                                                                                   |    |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----|
|    | <b>Frame engineering problems for ML</b><br>Turn a civil engineering question into inputs, features and a target. | 01 |
|    | <b>Wrangle real engineering data</b><br>Explore, clean and prepare datasets with Python, NumPy and pandas.        | 02 |
|  | <b>Predict engineering quantities</b><br>Build regression models for strength, response and performance.          | 03 |
|  | <b>Support engineering decisions</b><br>Build classifiers and judge them with precision, recall and F1.           | 04 |
|  | <b>Choose the right model</b><br>Compare linear, tree-based, ensemble and neural-network approaches.              | 05 |
|  | <b>Validate like an engineer</b><br>Catch data leakage, overfitting and unsafe extrapolation.                     | 06 |
|  | <b>Explain what the model learned</b><br>Use feature importance and SHAP to test physical plausibility.           | 07 |
|  | <b>Quantify uncertainty</b><br>Estimate probability of failure with Monte Carlo and ML surrogates.                | 08 |

# Sessions 01–04: the foundations.

Python, data preparation, regression and nonlinear models, on real civil engineering data.

01

## AI & ML for civil engineers: from problems to Python

 **Case study** · Concrete compressive strength

What AI, ML and deep learning really mean for engineers, then your first steps in Colab.

AI/ML concepts

Google Colab

Python

NumPy

pandas

First plots

02

## From raw engineering data to an ML-ready dataset

 **Case study** · Concrete + cross-domain engineering data

Learn to question a dataset before you model it.

EDA

Data quality

Outliers

Correlations

Train/test split

Scaling

Leakage

03

## Your first engineering prediction model

 **Case study** · Concrete compressive strength

Fit, score and diagnose a regression model the right way.

Linear regression

MAE · MSE · RMSE

$R^2$

Residuals

Cross-validation

04

## Nonlinear engineering problems: trees & ensembles

 **Case study** · Concrete + geotechnical response

When straight lines fail: models that capture nonlinear behaviour.

Decision trees

Random Forest

Boosting

Model comparison



**Milestone:** by session 04 you will have built, validated and compared linear, tree and ensemble models on real concrete data.

# Sessions 05–08: prediction to decision.

05

## From prediction to engineering decision: classification

 **Case study** · Environmental / water + geotechnical classification

Move from “how much?” to “safe or unsafe?”.

Logistic regression

Decision trees

Confusion matrix

Precision · Recall · F1

06

## Artificial neural networks: from neurons to prediction

 **Case study** · Concrete & engineering response prediction

Open the black box: how a network learns, and when it overfits.

ANN architecture

Activation

Loss

Epochs

Overfitting

Convergence

07

## Can we trust our model? Explainability & judgement

 **RESEARCH AREA**

 **Case study** · Geotechnical research case study

Check whether a model’s reasoning makes engineering sense.

Feature importance

Permutation importance

SHAP

Extrapolation

Physical plausibility

08

## From prediction to probability: uncertainty quantification

 **RESEARCH AREA**

 **Case study** · Reliability / engineering exceedance

Why a single number is not enough, and what to report instead.

Monte Carlo simulation

Surrogate models

Probability of failure

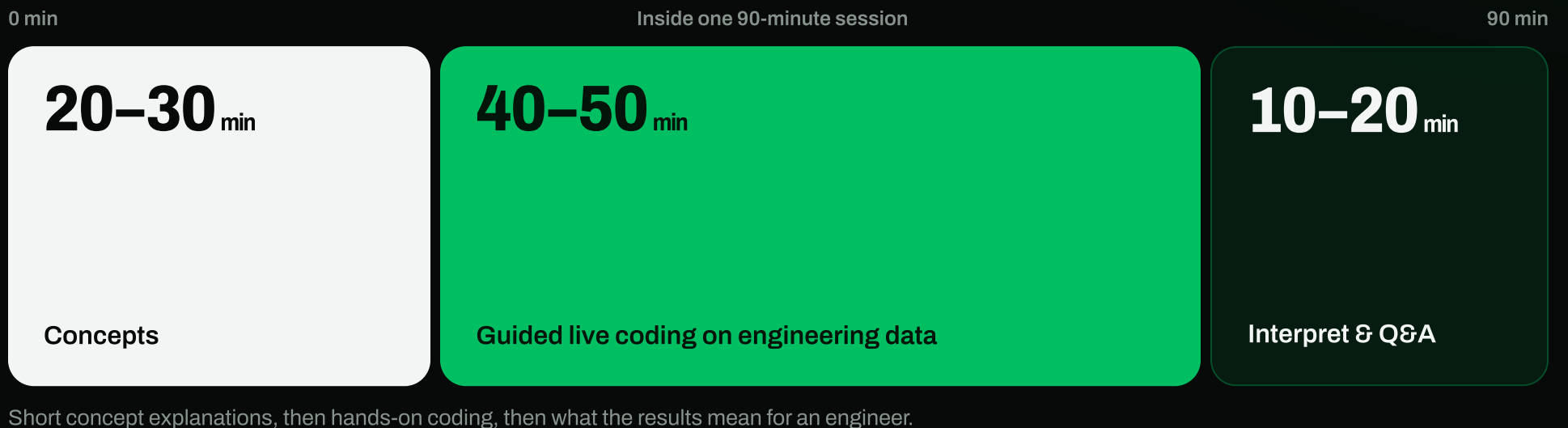
Uncertainty



**Beyond a typical ML course:** sessions 07 and 08 draw directly on the instructor’s research in explainable ML and reliability analysis.

# Fewer slides. More code.

Every 90-minute session is live and instructor-led, and roughly half of it is spent writing code together.



 **Live instructor-led coding**  
Code is written live with you, not pasted from slides.

 **Step-by-step notebooks**  
Every step explained in plain language inside the notebook.

 **Worked example, then your turn**  
Guided exercises follow every demonstration.

**Assignments with solutions**  
Practise on your own, then compare with the instructor's.

 **Engineering discussion**  
Ask questions and debate what the results really mean.

 **Recordings for revision**  
Missed a session? Catch up for up to a month.

 **Engineering judgement is built in.** Every session discusses model limitations, validation, extrapolation and physical plausibility, not just accuracy.

# Real engineering data. Right in your browser.

 03\_first\_prediction\_model.ipynb

● Google Colab · Python 3

```
[1] import pandas as pd
df = pd.read_csv("concrete_strength.csv")
df.shape

(1030, 9)
```

```
[2] from sklearn.model_selection import train_test_split
X, y = df.drop(columns="strength_mpa"), df["strength_mpa"]
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2)
```

```
[3] from sklearn.linear_model import LinearRegression
model = LinearRegression().fit(X_tr, y_tr) # your first model
```

## ZERO INSTALLS

- ✔ Works on any laptop or Chromebook

# Python

## NumPy

pandas

## Matplotlib

scikit-learn

SciPy

## PRIMARY DATASET

## UCI Concrete Compressive Strength

**1,030**  
concrete mixes

8  
input variables

0 missing values

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## WHAT GOES IN

## Cement

## Blast-furnace slag

## Fly ash

## Water

## Superplasticizer

## Coarse aggregate

Fine aggregate

Age (days)

↓ What the model predicts

### Compressive strength (MPa)

Plus curated, licence-checked examples from geotechnical, water, transport and building-energy engineering.

# Dr. Abdul Waris

**Research Associate**  
Universität Siegen (University of Siegen)  
Former Prime Minister’s Research Fellow · Ph.D., IIT Hyderabad

A computational geomechanics researcher who uses machine learning, reliability analysis and uncertainty quantification to make engineering decisions safer. His Ph.D. developed data-driven frameworks for the reliability analysis of slopes under rainfall.

- Scientific ML
- Explainable AI
- Surrogate modelling
- Reliability analysis
- Uncertainty quantification
- Offshore geotechnics



ACADEMIC JOURNEY

2016 – 2020

**B.Tech, Civil Engineering**

IUST, Awantipora

Ranked 2nd in a batch of 120+

2020 – 2022

**M.Tech, Geotechnical Engg.**

NIT Durgapur

Institute Gold Medalist

2022 – 2026

**Ph.D., Geotechnical Engg.**

IIT Hyderabad

PM Research Fellow · Research Excellence Award 2026

2026 – NOW

**Research Associate**

Universität Siegen, Germany

Teaches & supervises Master’s students

- WORKS WITH
- Python
- MATLAB
- PLAXIS
- ABAQUS
- HYDRUS
- GIS
- AutoCAD

# Taught by someone who publishes this work.

|                              |                       |                                   |                                  |
|------------------------------|-----------------------|-----------------------------------|----------------------------------|
| 10                           | 8                     | 9                                 | 4                                |
| international journal papers | in Q1-ranked journals | conference papers & book chapters | countries where he has presented |

PUBLISHED WITH    **ASCE**    **Springer**    **Elsevier**    **Nature Portfolio**

## Selected ML publications

**Explainable ensemble learning for estimating strength and stiffness of clay-bearing rocks**

Scientific Reports, Nature Portfolio · 2026

**Reliability analysis of unsaturated soil slopes using advanced surrogate models**

Journal of Computing in Civil Engineering, ASCE · in press

**Detection and prediction of slope stability using interpretable machine learning**

Modeling Earth Systems and Environment, Springer · 2025

**Pseudo-static slope stability analysis using explainable ML techniques**

Journal of Natural Hazards, Springer · 2024

## Invited lectures & talks

**Machine Learning for Geotechnical Engineers: Fundamentals & Practical Applications**

**Oregon State University, USA** · Invited talk · 2024

**Five-day workshop: ML & Finite Element Methods in Geotechnical Engineering**

**NIT Srinagar** · Resource person · 2025

**Unlocking Geotechnical Insights with Machine Learning**

**D.Y. Patil Institute of Technology, Pune** · National FDP · 2025

### Conference presentations

Oslo, Norway · Barcelona, Spain · 2025

## Teaching experience



### Master’s teaching & supervision

Universität Siegen, Germany



### Course instructor, Structural Analysis

B.Tech students, University of Kashmir



### Teaching assistant, Geotechnical Engg.

IIT Hyderabad & NIT Srinagar

# Everything you need, included.

No software to buy and nothing to install. Every notebook opens straight in Google Colab.



**8 teaching notebooks**  
Structured Colab notebooks, one per session



**Curated datasets**  
Cleaned, with data dictionaries



**Exercises & assignments**  
Practice problems for every topic



**Solution notebooks**  
Full instructor solutions to compare



**Python & ML quick reference**  
A cheat sheet to keep at your desk



**Session recordings**  
One month of access for revision

COHORT FALL 2026 · LIMITED SEATS

# Starts Saturday, 17 October 2026

 8 live sessions

 90 min each

 4 weeks

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